

Statistical Foundations for a Google Play User-Review Sentiment Index: Signal Fusion, Shrinkage, Distributional Validation, and Dynamic Smoothing

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Abstract

We develop a statistically explicit sentiment index for Google Play user reviews and establish the mathematical results supporting its construction. Normalized star ratings and text-sentiment scores are treated as noisy measures of latent review valence and fused by covariance-aware inverse-variance weighting. Review-level estimates are aggregated with bounded helpfulness and recency weights, then shrunk toward a population mean using estimated precision rather than an arbitrary review-count threshold. App-level rating histograms provide a distributional diagnostic for samples returned under different API sort orders; because star ratings are discrete, classical continuous Kolmogorov–Smirnov critical values are not used. A local-level state-space model and the Kalman filter provide a denoised temporal trend. Full proofs cover the BLUE and Gaussian maximum-likelihood result, Gaussian-conjugate shrinkage, the Glivenko–Cantelli and Donsker theorems, count transformations via the delta method, and exact Gaussian Kalman filtering. A worked three-review example shows how textual complaints can materially reduce an apparently perfect star-only score.

Keywords: sentiment index, online reviews, Google Play, inverse-variance weighting, empirical Bayes shrinkage, distributional validation, Kalman filter

MSC Classification: 62F10 , 62F15 , 60F05 , 60F17 , 62M20 , 93E11

1 Introduction

Online app reviews provide several imperfect signals of user experience: a one-to-five-star rating, free-form text, helpfulness counts, timestamps, app versions, and developer replies. A useful index must combine these signals without treating popularity as certainty, must remain stable in low-volume windows, and must distinguish temporal change from sampling artifacts. Sentiment analysis supplies review-level text scores [1–3]; the remaining problem is statistical aggregation.

This manuscript develops that aggregation framework and presents five self-contained supporting proofs: (i) inverse-variance weighting for combining star and text signals; (ii) Gaussian-conjugate shrinkage for low-information windows; (iii) uniform laws and weak convergence for distributional reasoning; (iv) the delta method for transformed count weights; and (v) Kalman filtering for a latent sentiment trend. Each proof makes its assumptions explicit. The final section maps the results to fields exposed by the `google-play-scraper` Python package, corrects common misuses of helpfulness counts and the Kolmogorov–Smirnov statistic, and gives a worked interpretation. Standard mathematical background is provided by [4–12].

2 Inverse-variance weighting is BLUE, and coincides with the Gaussian MLE

Setup.

Let $\theta \in \mathbb{R}$ be a fixed unknown parameter, and let

$$X_1 = \theta + \varepsilon_1, \quad X_2 = \theta + \varepsilon_2 \tag{1}$$

be two unbiased estimators satisfying $\mathbb{E}[\varepsilon_i] = 0$, $\text{Var}(\varepsilon_i) = \sigma_i^2$, and $\varepsilon_1 \perp \varepsilon_2$. Consider the class of linear combinations:

$$\hat{\theta}(w) = wX_1 + (1 - w)X_2, \quad w \in \mathbb{R}. \tag{2}$$

Step 1: Unbiasedness is automatic.

$$\mathbb{E}[\hat{\theta}(w)] = w\theta + (1 - w)\theta = \theta, \quad \forall w \in \mathbb{R}. \tag{3}$$

Thus, the entire parameterized family is unbiased; minimizing variance is the only remaining objective.

Step 2: Minimize variance.

By independence of ε_1 and ε_2 ,

$$V(w) := \text{Var}(\hat{\theta}(w)) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2. \tag{4}$$

This is a quadratic function in w with positive leading coefficient $\sigma_1^2 + \sigma_2^2 > 0$ (strictly convex). Hence, it admits a unique global minimum where $V'(w) = 0$:

$$\begin{aligned} V'(w) &= 2w\sigma_1^2 - 2(1-w)\sigma_2^2 = 0, \\ w^*(\sigma_1^2 + \sigma_2^2) &= \sigma_2^2, \\ w^* &= \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \frac{\sigma_1^{-2}}{\sigma_1^{-2} + \sigma_2^{-2}}. \end{aligned} \tag{5}$$

Step 3: Resulting minimum variance.

Substituting w^* back, with $a = \sigma_1^{-2}$ and $b = \sigma_2^{-2}$:

$$V(w^*) = \left(\frac{a}{a+b}\right)^2 \frac{1}{a} + \left(\frac{b}{a+b}\right)^2 \frac{1}{b} = \frac{a}{(a+b)^2} + \frac{b}{(a+b)^2} = \frac{1}{a+b} = \frac{1}{\sigma_1^{-2} + \sigma_2^{-2}}. \tag{6}$$

Consequently,

$$\hat{\theta}^* = w^* X_1 + (1 - w^*) X_2 = \frac{\sigma_1^{-2} X_1 + \sigma_2^{-2} X_2}{\sigma_1^{-2} + \sigma_2^{-2}} \tag{7}$$

is the unique variance-minimizing member of the linear unbiased class, i.e., the best linear unbiased estimator (BLUE). This is the two-source special case of Aitken's generalized least squares theorem.

Equivalence to the Gaussian MLE and UMVUE optimality.

Suppose also that $\varepsilon_i \sim \mathcal{N}(0, \sigma_i^2)$. The log-likelihood function is

$$\ell(\theta) = -\frac{(x_1 - \theta)^2}{2\sigma_1^2} - \frac{(x_2 - \theta)^2}{2\sigma_2^2} + \text{const}. \tag{8}$$

Setting $\ell'(\theta) = 0$:

$$\frac{x_1 - \theta}{\sigma_1^2} + \frac{x_2 - \theta}{\sigma_2^2} = 0 \implies \hat{\theta}_{\text{MLE}} = \frac{x_1 \sigma_1^{-2} + x_2 \sigma_2^{-2}}{\sigma_1^{-2} + \sigma_2^{-2}}, \tag{9}$$

which is identical to $\hat{\theta}^*$. Evaluating the second derivative gives $\ell''(\theta) = -(\sigma_1^{-2} + \sigma_2^{-2}) < 0$, confirming a global maximum.

Fisher information is additive over independent observations:

$$I(\theta) = \sigma_1^{-2} + \sigma_2^{-2}. \tag{10}$$

Since $\text{Var}(\hat{\theta}^*) = 1/I(\theta)$, the Cramér–Rao lower bound is achieved with exact equality. Therefore, $\hat{\theta}^*$ is BLUE and uniformly minimum-variance unbiased (UMVUE) among *all* unbiased estimators (linear or non-linear) in the Gaussian setting.

3 The shrinkage index is the Bayes posterior mean

Setup.

Let $I \mid \mu \sim \mathcal{N}(\mu, 1/V)$ denote the raw weighted index, treated as unbiased for the latent trend μ with precision V . Consider the prior $\mu \sim \mathcal{N}(C, 1/m)$.

Step 1: Posterior derivation via completing the square.

The posterior density satisfies:

$$p(\mu \mid I) \propto \exp\left[-\frac{1}{2}V(I - \mu)^2\right] \exp\left[-\frac{1}{2}m(\mu - C)^2\right]. \quad (11)$$

Expanding the exponent:

$$-\frac{1}{2}\left[(V + m)\mu^2 - 2(VI + mC)\mu + (VI^2 + mC^2)\right]. \quad (12)$$

Completing the square in μ :

$$(V + m)\mu^2 - 2(VI + mC)\mu = (V + m)\left[\mu - \frac{VI + mC}{V + m}\right]^2 - \frac{(VI + mC)^2}{V + m}. \quad (13)$$

Because the remainder does not depend on μ , it is absorbed into the normalization constant. Hence:

$$\mu \mid I \sim \mathcal{N}\left(\frac{VI + mC}{V + m}, \frac{1}{V + m}\right) = \mathcal{N}(I_{\text{shrunk}}, 1/(V + m)). \quad (14)$$

Precisions add directly ($V + m$), reflecting Gaussian-conjugate information pooling.

Step 2: Posterior mean as Bayes estimator under quadratic loss.

For any posterior $p(\mu \mid I)$ and any arbitrary candidate constant $c \in \mathbb{R}$:

$$\mathbb{E}[(\mu - c)^2 \mid I] = \mathbb{E}[(\mu - \mathbb{E}[\mu \mid I])^2 \mid I] + (\mathbb{E}[\mu \mid I] - c)^2, \quad (15)$$

since the cross-product term

$$2\mathbb{E}[(\mu - \mathbb{E}[\mu \mid I])(\mathbb{E}[\mu \mid I] - c) \mid I] = 2(\mathbb{E}[\mu \mid I] - c)\mathbb{E}[\mu - \mathbb{E}[\mu \mid I] \mid I] = 0. \quad (16)$$

The squared deviation is uniquely minimized at $c = \mathbb{E}[\mu \mid I]$. Hence $I_{\text{shrunk}} = \mathbb{E}[\mu \mid I]$ is the optimal Bayes estimator.

Step 3: Exact frequentist risk-dominance condition.

Let $a := V/(V + m)$, so that $I_{\text{shrunk}} = aI + (1 - a)C$. Evaluating under the true fixed parameter μ :

$$\text{Bias} = \mathbb{E}[I_{\text{shrunk}}] - \mu = a\mu + (1 - a)C - \mu = -(1 - a)(\mu - C), \quad (17)$$

$$\text{Var}(I_{\text{shrunk}}) = \frac{a^2}{V}. \quad (18)$$

The mean squared errors are:

$$\text{MSE}(I_{\text{shrunk}}) = (1-a)^2(\mu - C)^2 + \frac{a^2}{V}, \quad \text{MSE}(I) = \frac{1}{V}. \quad (19)$$

Demanding $\text{MSE}(I_{\text{shrunk}}) < \text{MSE}(I)$:

$$(1-a)^2(\mu - C)^2 < \frac{1-a^2}{V} = \frac{(1-a)(1+a)}{V} \implies (\mu - C)^2 < \frac{1+a}{V(1-a)}. \quad (20)$$

Substituting $a = V/(V+m)$, where $1-a = m/(V+m)$ and $1+a = (2V+m)/(V+m)$:

$$\boxed{(\mu - C)^2 < \frac{2V+m}{Vm}} \quad (21)$$

The boxed inequality is the exact boundary for MSE dominance. James–Stein shrinkage attains uniform risk dominance over the MLE for $p \geq 3$; in one dimension, no shrinkage estimator dominates uniformly over $\mu \in \mathbb{R}$. Risk degrades relative to the raw index whenever μ lies outside this critical radius, so the 1-D shrinkage estimator is Bayes-optimal under the specified prior but not uniformly dominant in a frequentist sense.

4 Glivenko–Cantelli theorem and Donsker’s theorem

4.1 Glivenko–Cantelli theorem

Theorem 4.1 (Glivenko–Cantelli) *Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$. The empirical distribution function $\hat{F}_n(x) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\}$ satisfies:*

$$D_n := \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \xrightarrow{\text{a.s.}} 0 \quad \text{as } n \rightarrow \infty. \quad (22)$$

Proof Step 1 (Pointwise SLLN). Fix $x \in \mathbb{R}$. The random variables $\mathbf{1}\{X_i \leq x\}$ are i.i.d. Bernoulli($F(x)$). By the strong law of large numbers (SLLN), $\hat{F}_n(x) \xrightarrow{\text{a.s.}} F(x)$. By an identical argument on left limits, $\hat{F}_n(x^-) \xrightarrow{\text{a.s.}} F(x^-)$.

Step 2 (Discretization onto a finite quantile grid). Fix $k \in \mathbb{N}^+$. Define the grid:

$$x_j := \inf\{x \in \mathbb{R} : F(x) \geq j/k\}, \quad j = 1, \dots, k-1, \quad (23)$$

with boundary conditions $x_0 = -\infty$ and $x_k = +\infty$. By definition of the generalized inverse, $F(x_j^-) \leq j/k \leq F(x_j)$. On any partition interval $[x_{j-1}, x_j)$, the measure satisfies:

$$F(x_j^-) - F(x_{j-1}) \leq \frac{j}{k} - \frac{j-1}{k} = \frac{1}{k}. \quad (24)$$

Step 3 (Sandwich argument via monotonicity). For any $x \in [x_{j-1}, x_j]$, monotonicity of $\hat{F}_n$ and F yields:

$$\begin{aligned}\hat{F}_n(x) - F(x) &\leq \hat{F}_n(x_j^-) - F(x_{j-1}) \\ &= [\hat{F}_n(x_j^-) - F(x_j^-)] + [F(x_j^-) - F(x_{j-1})] \\ &\leq [\hat{F}_n(x_j^-) - F(x_j^-)] + \frac{1}{k}.\end{aligned}\tag{25}$$

Analogously, from the lower bound:

$$\hat{F}_n(x) - F(x) \geq \hat{F}_n(x_{j-1}) - F(x_j^-) \geq [\hat{F}_n(x_{j-1}) - F(x_{j-1})] - \frac{1}{k}.\tag{26}$$

Taking the absolute supremum over all $x \in \mathbb{R}$:

$$\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \leq \max_{0 \leq j \leq k} |\hat{F}_n(x_j) - F(x_j)| + \max_{0 \leq j \leq k} |\hat{F}_n(x_j^-) - F(x_j^-)| + \frac{1}{k}.\tag{27}$$

Step 4 (Almost sure convergence). The bound involves a maximum over $2(k+1)$ terms, each converging to zero almost surely by Step 1. Because a finite union of null sets has probability zero:

$$\limsup_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \leq \frac{1}{k} \quad \text{a.s.}\tag{28}$$

Since $k \in \mathbb{N}^+$ was arbitrary, taking $k \rightarrow \infty$ establishes $D_n \xrightarrow{\text{a.s.}} 0$. $\square$

4.2 Donsker's theorem

Theorem 4.2 (Donsker) *Let $X_1, X_2, \dots$ be i.i.d. with a continuous distribution function F , and let $F^{-1}(t) := \inf\{x : F(x) \geq t\}$ denote the quantile function (with the usual conventions at $t = 0, 1$). The empirical process*

$$\mathbb{G}_n(t) := \sqrt{n}(\hat{F}_n(F^{-1}(t)) - t), \quad t \in [0, 1],\tag{29}$$

converges weakly in $D[0, 1]$ equipped with the Skorokhod topology to a standard Brownian bridge B :

$$\mathbb{G}_n \Rightarrow B.\tag{30}$$

Proof Part 1: Finite-dimensional convergence. Fix $0 \leq t_1 < t_2 < \dots < t_k \leq 1$. Because F is continuous, $F \circ F^{-1}$ is the identity on $[0, 1]$ [10, Proposition 1] and $U_i = F(X_i)$ is Uniform(0, 1) [10, Proposition 2]. For each t , the events $\{U_i \leq t\}$ and $\{X_i \leq F^{-1}(t)\}$ differ only on $\{F(X_i) = t\}$, an event of probability zero; hence, almost surely,

$$\mathbb{G}_n(t_l) = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{1}\{U_i \leq t_l\} - t_l \right).\tag{31}$$

Define the random vector:

$$\mathbf{Y}_i = (\mathbf{1}\{U_i \leq t_1\}, \dots, \mathbf{1}\{U_i \leq t_k\})^\top \in \mathbb{R}^k.\tag{32}$$

The vectors $\{\mathbf{Y}_i\}_{i=1}^n$ are i.i.d. with expectation $\mathbb{E}[\mathbf{Y}_i] = (t_1, \dots, t_k)^\top$ and covariance matrix:

$$\Sigma_{lm} = \text{Cov}(\mathbf{1}\{U \leq t_l\}, \mathbf{1}\{U \leq t_m\}) = \mathbb{P}(U \leq \min(t_l, t_m)) - t_l t_m = \min(t_l, t_m) - t_l t_m.\tag{33}$$

By the multivariate central limit theorem (via the Cramér–Wold device):

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{Y}_i - (t_1, \dots, t_k)^\top \right) \Rightarrow \mathcal{N}(0, \Sigma).\tag{34}$$

Because the l -th coordinate matches $\mathbb{G}_n(t_l)$, this confirms:

$$(\mathbb{G}_n(t_1), \dots, \mathbb{G}_n(t_k))^\top \Rightarrow \mathcal{N}(0, \Sigma). \quad (35)$$

Since $\text{Cov}(B(s), B(t)) = \min(s, t) - st$, the limiting Gaussian vector matches the finite-dimensional distributions of the Brownian bridge.

Part 2: Tightness via Billingsley's product-moment bound. To upgrade finite-dimensional convergence to weak convergence in $D[0, 1]$, we verify the moment version of Billingsley's two-interval condition: there exist $\gamma > 0$, $\alpha > 1$, and a continuous non-decreasing function F on $[0, 1]$ (Billingsley's modulus function; unrelated to the distribution function of Theorem 4.2) such that

$$\mathbb{E}[|\mathbb{G}_n(t) - \mathbb{G}_n(s)|^\gamma |\mathbb{G}_n(u) - \mathbb{G}_n(t)|^\gamma] \leq (F(u) - F(s))^\alpha, \quad \forall 0 \leq s \leq t \leq u \leq 1. \quad (36)$$

Set $\gamma = 2$, $\alpha = 2$, and $F(t) = t$. The increments over the disjoint intervals $(s, t]$ and $(t, u]$ are scaled sums of centered indicators:

$$\Delta_1 := \mathbb{G}_n(t) - \mathbb{G}_n(s) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i, \quad \Delta_2 := \mathbb{G}_n(u) - \mathbb{G}_n(t) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \eta_i, \quad (37)$$

where, in the uniform coordinates of Part 1,

$$\xi_i = \mathbf{1}\{s < U_i \leq t\} - a, \quad \eta_i = \mathbf{1}\{t < U_i \leq u\} - b, \quad a := t - s, \quad b := u - t. \quad (38)$$

The two increments are *not* independent: within a single observation, $\mathbb{E}[\xi_i \eta_i] = -ab$. We therefore expand the product moment exactly. Using independence across observations, $n^2 \mathbb{E}[\Delta_1^2 \Delta_2^2] = \sum_{i,j,k,l} \mathbb{E}[\xi_i \xi_j \eta_k \eta_l]$, and every index pattern in which some observation carries a single ξ - or η -factor vanishes, because $\mathbb{E}[\xi_i] = \mathbb{E}[\eta_i] = 0$. The surviving patterns are

$$\begin{aligned} \text{(i) } i = j = k = l : & \quad n \text{ terms, } \quad \gamma_0 := \mathbb{E}[\xi_i^2 \eta_i^2] = ab(a + b - 3ab), \\ \text{(ii) } i = j \neq k = l : & \quad n(n-1) \text{ terms, } \quad \mathbb{E}[\xi_i^2] \mathbb{E}[\eta_k^2] = a(1-a)b(1-b), \\ \text{(iii) } \{i, k\} = \{j, l\}, i \neq j : & \quad 2n(n-1) \text{ terms, } \quad \mathbb{E}[\xi_i \eta_i] \mathbb{E}[\xi_j \eta_j] = a^2 b^2, \end{aligned} \quad (39)$$

where γ_0 follows by integrating $\xi^2 \eta^2$ over the three regions $(s, t]$, $(t, u]$, and the rest, with probabilities a , b , $1 - a - b$. Hence, exactly,

$$\mathbb{E}[\Delta_1^2 \Delta_2^2] = \frac{\gamma_0}{n} + \frac{n-1}{n} [a(1-a)b(1-b) + 2a^2 b^2]. \quad (40)$$

Both $\gamma_0 \leq ab$ (since $a + b \leq 1$) and $a(1-a)b(1-b) + 2a^2 b^2 = ab[1 - (a+b) + 3ab] \leq ab$ (since $3ab \leq a + b$ for $a + b \leq 1$); the right-hand side of (40) is a convex combination of two quantities bounded by ab , so

$$\mathbb{E}[\Delta_1^2 \Delta_2^2] \leq ab = (t-s)(u-t) \leq \frac{(u-s)^2}{4} \leq (F(u) - F(s))^\alpha, \quad (41)$$

by the arithmetic mean–geometric mean inequality, with $F(t) = t$ and $\alpha = 2$. This is (36) for $\gamma = 2$, $\alpha = 2$. By Markov's inequality, $\mathbb{P}\{|\Delta_1| > \lambda, |\Delta_2| > \lambda\} \leq \mathbb{E}[\Delta_1^2 \Delta_2^2] / \lambda^4 \leq (u-s)^2 / \lambda^4$, i.e. exactly condition (15.21) of [7] with $\alpha = 2$ and $\gamma = 4$. The finite-dimensional limits are the Brownian bridge, which has continuous paths, so $T_B = [0, 1]$, $\mathbb{P}\{B(1) \neq B(1-)\} = 0$, and the tightness/modulus conclusion of Billingsley's Theorem 15.6 follows from this bound via his maximal inequality (Theorem 12.5 and (15.22)). The same product-moment inequality, (40), is precisely the estimate (13.17)–(13.18) used in the proof of Billingsley's Theorem 13.1 and reused in the proof of his Theorem 16.4 for the $D[0, 1]$ -valued empirical process.

Part 3: Weak convergence and distribution-freeness. Finite-dimensional convergence (Part 1) and the two-interval bound (Part 2) combine to give $\mathbb{G}_n \Rightarrow B$ in $D[0, 1]$ by Billingsley's Theorem 15.6; the bound supplies the modulus estimate (15.22), hence tightness,

and Prokhorov's theorem applies. The functional $h(x) = \sup_{t \in [0,1]} |x(t)|$ is continuous at every continuous path in $D[0,1]$, and B has continuous paths almost surely. The continuous mapping theorem therefore gives:

$$\sup_{t \in [0,1]} |\mathbb{G}_n(t)| \Rightarrow \sup_{t \in [0,1]} |B(t)|. \quad (42)$$

The limiting law does not depend on F , which is why Kolmogorov–Smirnov critical values apply to any continuous distribution. $\square$

Remark 1 The continuity hypothesis on F in Theorem 4.2 is essential. For arbitrary F , both steps fail: (i) $U_i = F(X_i)$ is stochastically larger than uniform, $\mathbb{P}\{F(X) < t\} \leq t$ with strict inequality whenever t is outside the closure of the range of F [10, Proposition 2, (25)]; and (ii) $F(F^{-1}(t)) \geq t$ [10, Proposition 1, (24)], so that

$$\mathbb{E} \mathbb{G}_n(t) = \sqrt{n}(F(F^{-1}(t)) - t), \quad (43)$$

which diverges to $+\infty$ as $n \rightarrow \infty$ at every $t \in (0,1)$ outside the range of F , i.e. at every gap created by an atom of F , and $\mathbb{G}_n(t)$ itself then diverges in probability. The correct general- F statements are the x -indexed process $\sqrt{n}(\hat{F}_n(x) - F(x))$, which reduces pathwise, simultaneously for all n , to the uniform case through the quantile representation $\hat{F}_n = G_n \circ F$ [10, Theorem 3, (20)] and converges to the Brownian bridge with covariance $\mathbb{E}[B(s)B(t)] = F(s)(1 - F(t))$, $s \leq t$ [7, Theorem 16.4]; or the coupling formulation for uniform samples [11, Theorem 1.7]. For continuous F , $F \circ F^{-1} = \text{id}$ [10, Proposition 1], which is exactly what Parts 1–2 use.

5 The delta method and analysis of $\log(1 + U)$

Theorem 5.1 (Univariate delta method) *Let $\{X_n\}$ be a sequence of random variables such that $\sqrt{n}(X_n - \theta) \Rightarrow \mathcal{N}(0, \sigma^2)$. If g is differentiable at θ with $g'(\theta) \neq 0$, then:*

$$\sqrt{n}(g(X_n) - g(\theta)) \Rightarrow \mathcal{N}(0, [g'(\theta)]^2 \sigma^2). \quad (44)$$

Proof Taylor expanding g around θ :

$$g(x) = g(\theta) + g'(\theta)(x - \theta) + R(x), \quad \text{where } \lim_{x \rightarrow \theta} \frac{R(x)}{x - \theta} = 0. \quad (45)$$

Substituting X_n :

$$\sqrt{n}(g(X_n) - g(\theta)) = g'(\theta)\sqrt{n}(X_n - \theta) + \underbrace{\sqrt{n}(X_n - \theta)}_{O_p(1)} \cdot \underbrace{\frac{R(X_n)}{X_n - \theta}}_{o_p(1)}. \quad (46)$$

Since $X_n - \theta = O_p(n^{-1/2}) \xrightarrow{p} 0$, continuity ensures $R(X_n)/(X_n - \theta) \xrightarrow{p} 0$. By Slutsky's theorem, the remainder is $o_p(1)$, concluding:

$$\sqrt{n}(g(X_n) - g(\theta)) = g'(\theta)\sqrt{n}(X_n - \theta) + o_p(1) \Rightarrow \mathcal{N}(0, [g'(\theta)]^2 \sigma^2). \quad (47)$$

$\square$

Application to Poisson counts.

For $U \sim \text{Poisson}(\lambda)$ where $\text{Var}(U) = \lambda$, the first-order Taylor propagation of error yields:

$$\text{Var}(g(U)) \approx [g'(\lambda)]^2 \text{Var}(U) = [g'(\lambda)]^2 \lambda. \quad (48)$$

Setting this equal to a constant c :

$$g'(\lambda) = \sqrt{c} \lambda^{-1/2} \implies g(\lambda) = 2\sqrt{c\lambda} + C_0. \quad (49)$$

Thus, $g(\lambda) \propto \sqrt{\lambda}$ is the unique (up to affine transformation) variance-stabilizing transformation.

Asymptotic variance of $\log(1 + U)$.

For $g(\lambda) = \log(1 + \lambda)$, the first derivative is $g'(\lambda) = \frac{1}{1+\lambda}$, yielding asymptotic variance:

$$[g'(\lambda)]^2 \lambda = \frac{\lambda}{(1 + \lambda)^2} \xrightarrow{\lambda \rightarrow \infty} 0. \quad (50)$$

This variance decays to zero as λ grows, so $\log(1 + U)$ is an *over-stabilizing concave dampener*.

While the variance stabilizer for Poisson counts is the Anscombe transform $\sqrt{U + 3/8}$, the logarithmic transform $\log(1 + U)$ damps large-count outliers because its derivative decays as $\mathcal{O}(\lambda^{-1})$ rather than $\mathcal{O}(\lambda^{-1/2})$.

6 The Kalman filter for the local-level model

Consider the state-space system:

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \quad (51)$$

$$I_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (52)$$

with prior $\mu_0 \sim \mathcal{N}(m_0, P_0)$, and mutually independent noise sequences.

Lemma 6.1 (Gaussian conditional distributions) *Let $(X, Y)^\top$ be jointly Gaussian with Σ_{YY} invertible:*

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix} \right). \quad (53)$$

Then the conditional distribution is:

$$X | Y \sim \mathcal{N}(\mu_X + \Sigma_{XY} \Sigma_{YY}^{-1} (Y - \mu_Y), \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX}). \quad (54)$$

Proof Define the auxiliary variable $Z := X - \Sigma_{XY} \Sigma_{YY}^{-1} (Y - \mu_Y) - \mu_X$. Because Z is an affine combination of jointly Gaussian variables, $(Z, Y)^\top$ is jointly Gaussian. The cross-covariance is:

$$\text{Cov}(Z, Y) = \text{Cov}(X, Y) - \Sigma_{XY} \Sigma_{YY}^{-1} \text{Var}(Y) = \Sigma_{XY} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YY} = 0. \quad (55)$$

For jointly Gaussian vectors, uncorrelatedness implies independence: $Z \perp Y$. Therefore:

$$\mathbb{E}[Z | Y] = \mathbb{E}[Z] = 0, \quad (56)$$

$$\text{Var}(Z | Y) = \text{Var}(Z) = \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX}. \quad (57)$$

Re-substituting $X = Z + \mu_X + \Sigma_{XY} \Sigma_{YY}^{-1} (Y - \mu_Y)$ gives the stated conditional moments. $\square$

Exact MMSE optimality via induction.

Any linear transformation of Gaussian primitives remains Gaussian, so the joint vector $(\mu_t, I_{1:t})$ is Gaussian and every conditional expectation $\mathbb{E}[\mu_t | I_{1:t}]$ is linear in the observations. Because the conditional expectation $\mathbb{E}[X | Y]$ uniquely minimizes the mean squared error over *all* measurable functions (linear or non-linear), the linear Kalman recursion achieves exact Bayes MMSE optimality.

Inductive recursion.

- **Base case** ($t = 0$):

$$\hat{\mu}_{0|0} = m_0, \quad P_{0|0} = P_0. \quad (58)$$

- **Time update (predict):** Assume $\mu_{t-1} | I_{1:t-1} \sim \mathcal{N}(\hat{\mu}_{t-1|t-1}, P_{t-1|t-1})$. Since $\eta_t \perp I_{1:t-1}$:

$$\hat{\mu}_{t|t-1} = \mathbb{E}[\mu_{t-1} + \eta_t | I_{1:t-1}] = \hat{\mu}_{t-1|t-1}, \quad (59)$$

$$P_{t|t-1} = \text{Var}(\mu_{t-1} + \eta_t | I_{1:t-1}) = P_{t-1|t-1} + \tau^2. \quad (60)$$

- **Measurement update (correct):** Conditional on $I_{1:t-1}$, the vector $(\mu_t, I_t)^\top$ is jointly Gaussian:

$$\text{Cov}(\mu_t, I_t | I_{1:t-1}) = \text{Var}(\mu_t | I_{1:t-1}) = P_{t|t-1}, \quad (61)$$

$$\text{Var}(I_t | I_{1:t-1}) = P_{t|t-1} + \sigma^2 > 0. \quad (62)$$

Applying Lemma 6.1:

$$K_t := \frac{P_{t|t-1}}{P_{t|t-1} + \sigma^2}, \quad (63)$$

$$\hat{\mu}_{t|t} = \hat{\mu}_{t|t-1} + K_t (I_t - \hat{\mu}_{t|t-1}), \quad (64)$$

$$P_{t|t} = P_{t|t-1} - K_t P_{t|t-1} = \frac{P_{t|t-1} \sigma^2}{P_{t|t-1} + \sigma^2}. \quad (65)$$

This completes the induction: $\mu_t | I_{1:t} \sim \mathcal{N}(\hat{\mu}_{t|t}, P_{t|t})$ is the exact Bayes MMSE estimator.

7 A Google Play user-review sentiment index

This section converts the preceding results into an estimable index using the schema exposed by the `google-play-scraper` Python package [13]. The purpose is not to replace a trained sentiment model, but to specify how text scores, star ratings, review metadata, and app-level rating distributions should be combined and interpreted. The design also separates quantities observed through the scraper from assumptions that must be estimated or chosen by the analyst.

7.1 Observable fields and their statistical roles

The app-detail call exposes the app identifier, aggregate score, numbers of ratings and reviews, the five-star histogram, current version, and update time. The review call adds the review text, star score, helpfulness count, timestamp, reviewed app version, and any developer reply. Table 1 maps these fields to their roles in the proposed index.

Table 1 Google Play fields used in the sentiment-index pipeline.

Field	Statistical role	Use and limitation
content	Text-sentiment signal	Produces a polarity score and model uncertainty; performance depends on language, domain, and handling of negation or sarcasm.
score	Ordinal rating signal	Maps a one-to-five-star response to a common $[-1, 1]$ scale; it need not agree with the written review.
thumbsUpCount	Social endorsement	May modestly affect precision or prominence, but lacks exposure and down-vote denominators and is therefore not a Wilson proportion.
at	Review time	Supports recency weighting and time-window aggregation.
reviewCreatedVersion, appVersion	Release context	Supports version-level comparisons and change-point analysis.
replyContent, repliedAt	Developer response	Provides optional responsiveness covariates; a reply is not itself evidence that the complaint was resolved.
histogram	App-level rating distribution	Supplies a benchmark for diagnosing star-rating selection bias in the scraped review sample.

The review endpoint paginates results and permits alternative sort orders. Thus, a call returning the newest or most relevant reviews is a selected collection, not automatically a simple random sample. Retrieving all exposed reviews reduces truncation but can require many requests and still does not correct platform-level selection: only users who chose to rate or review the app are observed.

7.2 Review-level latent valence

Let $r_i \in \{1, 2, 3, 4, 5\}$ denote review i 's star rating. Map it to

$$\rho_i = \frac{r_i - 3}{2} \in [-1, 1]. \quad (66)$$

A lexicon method such as VADER or a validated statistical classifier produces a text score $s_i \in [-1, 1]$ [3]. Model-based class probabilities may be converted to a continuous score by taking the expected numerical class value. In either case, calibration should be checked on hand-labeled reviews from the target language and app domain.

Treat ρ_i and s_i as two noisy measures of a latent review valence θ_i . Define $\mathbf{y}_i = (\rho_i, s_i)^\top$ and suppose

$$\mathbf{y}_i = \theta_i \mathbf{1} + \mathbf{e}_i, \quad \mathbb{E}(\mathbf{e}_i) = \mathbf{0}, \quad \text{Var}(\mathbf{e}_i) = \mathbf{\Sigma}_i. \quad (67)$$

The covariance-aware best linear unbiased estimate is

$$\hat{\theta}_i = \frac{\mathbf{1}^\top \Sigma_i^{-1} \mathbf{y}_i}{\mathbf{1}^\top \Sigma_i^{-1} \mathbf{1}}, \quad \nu_i := \text{Var}(\hat{\theta}_i) = \frac{1}{\mathbf{1}^\top \Sigma_i^{-1} \mathbf{1}}. \quad (68)$$

When the two errors are uncorrelated, equation (68) reduces to the inverse-variance estimator proved in Section 2. The covariance matrix should be estimated from a labeled validation set; star ratings and text scores should not be assumed independent merely for convenience. If such data are unavailable, a prespecified convex combination is more transparent than a spurious precision estimate.

The disagreement statistic

$$\delta_i = |\rho_i - s_i| \quad (69)$$

is retained as a diagnostic. A large value can reveal rating-text mismatch, sarcasm, accidental star selection, or classifier failure. It is a flag for inspection, not proof that a review is deceptive.

7.3 Helpfulness, recency, and window aggregation

Raw helpfulness counts are highly exposure-dependent: older or prominently displayed reviews have had more opportunities to receive votes. A Wilson lower bound requires both successes and trials [14]; `thumbsUpCount` supplies neither total impressions nor negative votes. Consequently, helpfulness should have bounded influence. One practical weight is

$$w_i = q_i d_i, \quad q_i = 1 + \alpha \min\{\log(1 + u_i), q_{\max}\}, \quad d_i = \exp\{-\lambda(T - t_i)\}, \quad (70)$$

where u_i is the helpfulness count, $T - t_i$ is review age in a fixed time unit, $\alpha \geq 0$ controls social-endorsement influence, and $\lambda \geq 0$ controls decay. The leading 1 prevents reviews with zero helpfulness votes from receiving zero weight. The logarithm dampens the heavy right tail; Section 5 explains why its marginal effect declines rapidly.

For reviews in time or version window t , define

$$I_t = \frac{\sum_{i \in \mathcal{W}_t} w_i \hat{\theta}_i}{\sum_{i \in \mathcal{W}_t} w_i}. \quad (71)$$

If review-level errors are conditionally independent, a working variance that combines signal-fusion uncertainty with observed between-review heterogeneity is

$$S_t^2 = \frac{\sum_{i \in \mathcal{W}_t} w_i^2 \left\{ \nu_i + (\hat{\theta}_i - I_t)^2 \right\}}{(\sum_{i \in \mathcal{W}_t} w_i)^2}. \quad (72)$$

For very small windows, this variance is unstable and should be regularized or estimated by a prespecified bootstrap. A cluster-robust or campaign-level variance should replace equation (72) when reviews are dependent. The effective sample size

$$n_{\text{eff},t} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} \quad (73)$$

should accompany the nominal review count because a few highly weighted reviews can otherwise create false confidence.

7.4 Precision-based shrinkage and temporal smoothing

Let $P_t = 1/S_t^2$ be the estimated data precision, let C be a global or category-level reference mean, and let m be its prior precision. The shrunk window index is

$$I_t^B = \frac{P_t I_t + mC}{P_t + m}. \quad (74)$$

This is the Gaussian posterior mean proved in Section 3. Importantly, P_t is estimated precision, not simply the number of reviews or the sum of arbitrary weights. The prior parameters should be estimated from historical apps or declared in advance, and sensitivity to m should be reported.

For ordered windows, let the latent app sentiment follow the local-level model

$$I_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, S_t^2), \quad \mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \tau^2). \quad (75)$$

The Kalman recursion in Section 6 then yields a denoised estimate of μ_t . The raw index I_t , rather than the already shrunk I_t^B , enters the observation equation so that prior information is not counted twice. Static shrinkage is useful when windows are analyzed separately; in a time-series analysis, C and m can instead inform the initial state. Version indicators may be examined as interventions, but a change at a release boundary is associational unless competing time trends and changes in the reviewing population are addressed.

7.5 Distributional validation against the app histogram

Let p_k be the proportion of all app ratings in star category k according to the app-level histogram, and let $\hat{p}_k$ be the corresponding proportion in the scraped review sample. A useful discrepancy measure is

$$D_\star = \max_{1 \leq k \leq 5} \left| \sum_{j=1}^k \hat{p}_j - \sum_{j=1}^k p_j \right|. \quad (76)$$

Although equation (76) has the form of a Kolmogorov–Smirnov distance, star ratings are discrete. Therefore the standard continuous-distribution critical values associated with Section 4 are not valid. Under a defensible random-sampling assumption, uncertainty may instead be calibrated by a multinomial parametric bootstrap or a Pearson goodness-of-fit statistic. With API-ranked samples, the more honest use of $D_\star$ is descriptive: compare sort orders, collection dates, and version windows, and report how far the sample rating distribution is from the platform histogram.

When every star category is represented, post-stratification can align the sample with the app-level distribution by multiplying review weights in category k by $p_k/\hat{p}_k$. This corrects the observed star mix but cannot remove selection bias within a star category or recover a category absent from the sample.

7.6 Worked interpretation

Consider the three supplied Penguin Isle reviews. All have five stars, so $\rho_i = 1$, but their texts range from enthusiastic praise to a report that the app stopped opening. For illustration, suppose a calibrated fusion model assigns weight 0.625 to the star signal and 0.375 to the text signal. The text scores in Table 2 are illustrative inputs rather than newly fitted classifier outputs. Recency is held constant, and the displayed weight is $\log(1 + u_i)$ solely to reproduce the compact example; an operational system should use the positive, capped weight in equation (70).

Table 2 Illustrative fusion of three reviews.

Reviewer	Stars	Helpful	s_i	$\hat{\theta}_i$	$\log(1 + u_i)$
LiviaBrannock	5	239	0.30	0.737	5.48
MorganEllis	5	12	-0.50	0.438	2.57
AGoogleuser	5	116	0.90	0.963	4.76

The weighted raw index is

$$I = \frac{5.48(0.737) + 2.57(0.438) + 4.76(0.963)}{5.48 + 2.57 + 4.76} = 0.761. \quad (77)$$

The naive star-only index is 1.000, so text fusion exposes a complaint pattern hidden by the nominally perfect ratings. MorganEllis has the largest disagreement, $|1 - (-0.50)| = 1.50$, and its fused valence falls to 0.438. The result should still not be described as precise: there are only three reviews and the effective sample size is below three.

To illustrate shrinkage, take $C = 0.40$, $m = 50$, and a deliberately low assumed data precision $P = 12.8$. This precision is an illustrative input, not a quantity inferred from the review count or the displayed weights. Equation (74) gives

$$I^B = \frac{12.8(0.761) + 50(0.40)}{12.8 + 50} = 0.474. \quad (78)$$

The value may be described as moderately positive under a prespecified band such as 0.2 to 0.6, but the band is a reporting convention rather than a universal statistical threshold. The defensible interpretation is: the observed reviews lean positive, one review contains a substantial star-text mismatch, and the small sample does not justify the perfect sentiment implied by the star average. An empirical analysis should report I_t , I_t^B , an uncertainty interval, nominal and effective sample sizes, the rating-distribution discrepancy, the collection parameters (language, country, sort order, and date), and the app version represented by each window.

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